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Bidirectional Vietnamese Dialect Transfer

Region-Conditioned Standard-to-Dialect Generation

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Abstract

Vietnamese dialectal variation is prominent in everyday communication, especially in social-media text, but most existing NLP work treats it primarily as a source of noise to be normalized into standard Vietnamese. In this work, we study Vietnamese dialect transfer in both directions. We formulate region-conditioned standard-to-dialect generation as a complementary task to dialect-to-standard normalization, targeting the Northern, Central, and Southern dialect regions within a unified sequence-to-sequence framework. We fine-tune a single BARTpho-syllable model with task-prefix conditioning to perform three related mappings: dialect-to-standard normalization, lexical normalization, and region-conditioned standard-to-dialect generation. To build the generation data, we reverse the ViDia2Std dialect–standard pairs while retaining the original human-written dialect sentence as the target. Since the resulting data is strongly region-imbalanced, we further apply region rebalancing and sampling-based back-translation for the lower-resource regions. Because standard-to-dialect generation is inherently one-to-many, we evaluate it with both conventional overlap metrics and transfer-oriented metrics, including Dialect Feature Recall, Dialectal Edit-Recall, and copy-rate. The experiments show that the multi-task model remains competitive on normalization while learning meaningful dialectal substitutions in the reverse direction. The results also reveal persistent challenges, particularly for low-resource regions and for automatic evaluation with single-reference dialectal targets. Overall, this work provides a practical framework for bidirectional Vietnamese dialect transfer and establishes region-conditioned dialect generation as a measurable task for future study.

1 Introduction

Vietnamese is spoken by roughly one hundred million people across three broad dialect regions—Northern (centred on Ha Noi), Central (e.g. Nghe An, Ha Tinh, Hue), and Southern (centred on Ho Chi Minh City). These dialects differ in lexical choice, function words, pronouns, and orthographic realization of colloquial speech. A Central speaker may write *"Anh đi mô rồi?"* where a standard speaker writes *"Anh đi đâu rồi?"* ("Where did he go?"); the Central marker *mô* ("where") and the Southern *hông* / *hổng* ("not") have no place in formal written Vietnamese but are pervasive on social media.

Almost all prior computational treatment of this variation is *normalizing*: it maps dialectal or noisy text onto a standard target. The recent ViDia2Std benchmark (Ta et al., 2026) and the Arabic NADI shared tasks (Abdul-Mageed et al., 2024) both frame the problem as dialect→standard. The reverse direction—generating idiomatic regional text *from* a standard sentence—has, to our knowledge, no published treatment for Vietnamese. Yet this direction is valuable for data augmentation, for controllable and localized text generation, for stylistic personas in dialogue systems, and for studying the linguistic structure of the dialects themselves.

This report describes a capstone system that addresses both directions inside a *single* model. Our contributions are:

1. **A novel task: region-conditioned standard→dialect generation** (Feature C) for Northern, Central, and Southern Vietnamese, realized by reversing the ViDia2Std parallel corpus and conditioning a seq2seq model on a region-specific task prefix.
2. **A unified multi-task model**: one BARTpho-syllable model performs dialect→standard normalization (A), region-conditioned standard→dialect generation (C), and ViLexNorm lexical normalization (B), all selected by a textual task prefix.
3. **An evaluation suite appropriate to one-to-many generation**: we argue that single-reference BLEU under-rewards valid stylistic diversity (Edunov et al., 2018) and introduce Dialect Feature Recall (DFR), a marker-independent Dialectal Edit-Recall, and a marker-based region-accuracy measure, complemented by a human-evaluation protocol.
4. **Data strategy for severe regional imbalance**: oversampling plus sampling-based back-translation (Edunov et al., 2018) that keeps *real* dialect as the target and synthesizes standard sources, directed at the data-starved Northern and Southern regions.

On the well-studied normalization direction our syllable-level model is competitive with much larger systems (Section 8), which we present as a positioning result rather than a claim of state of the art.

2 Related Work

Vietnamese pretrained seq2seq models. Two encoder–decoder backbones dominate Vietnamese generation. BARTpho (Tran et al., 2022) adapts the BART/mBART denoising objective (Lewis et al., 2020; Liu et al., 2020) to Vietnamese and is released in two tokenization variants: *BARTpho-syllable*, which segments text at the syllable level (the natural orthographic unit of Vietnamese, since white space separates syllables rather than words), and *BARTpho-word*, which first applies automatic word segmentation. ViT5 (Phan et al., 2022) is a Vietnamese T5 (Raffel et al., 2020) trained with a span-corruption objective. Both are strong for normalization and summarization; we adopt BARTpho-syllable because syllable-level tokenization is robust to the out-of-vocabulary, non-standard spellings that pervade dialectal and social-media text, where a word segmenter trained on standard Vietnamese is unreliable. For dialect identification we note PhoBERT (Nguyen and Nguyen, 2020), a RoBERTa-style Vietnamese encoder, as a natural classifier backbone, and VnCoreNLP (Vu et al., 2018) for segmentation and POS analysis.

Dialect-to-standard normalization. ViDia2Std (Ta et al., 2026) is the first large parallel corpus of Vietnamese dialect/standard sentence pairs, collected from public Facebook comments across all 63 provinces and region-labelled by native annotators. The authors frame the task as monolingual normalization (dialect→standard) and benchmark a range of seq2seq models, with mBART-large-50 the strongest. The NADI shared tasks for Arabic (Abdul-Mageed et al., 2024) similarly target dialect identification and dialect→MSA normalization. ViLexNorm (Nguyen et al., 2024) addresses the adjacent problem of *lexical* normalization—mapping teencode and abbreviated social-media spellings to their standard forms—and VSEC (Do et al., 2021) addresses Vietnamese spelling correction. All of these are *many-to-one* normalization tasks. We incorporate the normalization direction (Features A and B) but argue that the inverse, generative direction is both unsolved and qualitatively different.

Text style transfer and dialect generation. The reverse direction is a form of text style transfer. Non-parallel methods learn to disentangle content from style with adversarial cross-alignment (Shen et al., 2017) or via unsupervised-MT machinery (Lample et al., 2018). Closest to our work, Le and Luu (2023) study Central↔Northern Vietnamese dialect text transfer, but restrict themselves to two regions and a pairwise transfer setting. We instead condition a single model on three regions and treat the standard variety as a shared pivot, so that generation and normalization share one parameter set and one tokenizer. To our knowledge no prior work performs region-conditioned standard→dialect generation across all three Vietnamese regions.

Back-translation and monolingual data. Back-translation (Sennrich et al., 2016b) augments a translation system by pairing real target-side sentences with machine-generated source-side sentences. Edunov et al. (2018) show that *sampling* (or noised-beam) back-translation outperforms pure beam search, because greedy/beam decoding produces overly regular synthetic sources that under-represent the true data distribution and induce mode collapse; they also caution that BLEU correlates poorly with the diversity and quality that back-translation actually adds. We follow this prescription directly: we keep *real* dialect sentences as targets and *sample* synthetic standard sources from a strong dialect→standard model, concentrating the augmentation on the data-starved Northern and Southern regions.

Evaluation of generation. Corpus BLEU (Papineni et al., 2002; Post, 2018), chrF (Popović, 2015), ROUGE-L (Lin, 2004), METEOR (Banerjee and Lavie, 2005), and the embedding-based BERTScore (Zhang et al., 2020) are the standard automatic metrics. None of them is designed to reward producing *a* valid dialectal realization among many; we therefore complement them with task-specific measures (Section 7).

3 Background and NLP Fundamentals

This section develops the modelling and evaluation concepts the system relies on, in the order they appear in the method.

3.1 Tokenization for Vietnamese: syllables vs. words vs. subwords

Written Vietnamese delimits *syllables* (*tiếng*) with white space; a "word" may comprise one or several syllables (e.g. *sinh viên*, "student", is two syllables but one word). A model can therefore tokenize at three granularities: (i) automatic *word* segmentation, as performed by VnCoreNLP (Vu et al., 2018) and used by BARTpho-word and PhoBERT (Nguyen and Nguyen, 2020); (ii) raw *syllables*, used by BARTpho-syllable (Tran et al., 2022); and (iii) statistical *subword* units such as BPE (Sennrich et al., 2016a) applied on top of either. Word segmentation injects useful structure on clean text, but it is brittle on dialectal and social-media input, where non-standard spellings (*không* → *hông*), elongations, and region-specific function words are out-of-vocabulary for a segmenter trained on standard Vietnamese. A segmentation error then propagates to every downstream token. Syllable-level tokenization sidesteps this: every white-space-delimited unit is a valid token, the vocabulary is small and stable, and rare dialectal forms degrade gracefully into shorter, still-meaningful units. For a task whose entire input distribution is dialectal/noisy text, this robustness is decisive, which is why we adopt BARTpho-syllable.

3.2 Conditional sequence-to-sequence generation

All three tasks are cast as conditional generation. Given a source token sequence $x = (x_1, \ldots, x_m)$ and a discrete *condition* c (here a task/region prefix), an encoder–decoder Transformer (Vaswani et al., 2017) models the target $y = (y_1, \ldots, y_n)$ autoregressively:

$$P_\theta(y \mid x, c) \;=\; \prod_{t=1}^{n} P_\theta(y_t \mid y_{<t},\, x,\, c)\,. \tag{1}$$

We realize the condition c *textually*, by prepending a task prefix to the source so that the input is the string "`{task}: {source}`". This keeps the architecture unchanged—no new embeddings or heads—and lets a single set of parameters serve every task, with the prefix steering generation.

3.3 Training objective: teacher forcing, cross-entropy, label smoothing

The model is trained by *teacher forcing*: at decoder step t the ground-truth prefix $y_{<t}$ is fed in regardless of the model's own predictions, which makes the per-step targets independent and the loss fully parallelizable over t. The objective is token-level cross-entropy (negative log-likelihood),

$$\mathcal{L}(\theta) \;=\; -\sum_{t=1}^{n} \log P_\theta(y_t \mid y_{<t}, x, c)\,. \tag{2}$$

Minimizing Eq. 2 drives the model toward a one-hot target distribution, which can over-confidently penalize the many *other* valid continuations that exist in a one-to-many task. *Label smoothing* (Szegedy et al., 2016) mixes the one-hot target with a uniform distribution of mass

ϵ, replacing the target probability of the gold token with $1 - \epsilon$ and distributing ϵ over the vocabulary. This regularizes the model, improves calibration, and is especially appropriate when several outputs are acceptable—as in standard→dialect generation.

3.4 Decoding: beam search vs. sampling

At inference we must search Eq. 1 for an output. *Beam search* keeps the k highest-probability partial hypotheses at each step and returns the most probable complete sequence; it is the right choice when the task is essentially deterministic (one correct answer), as in normalization. *Sampling* draws $y_t \sim P_\theta(\cdot \mid y_{<t}, x, c)$, optionally restricted to the top-k tokens or to the smallest set whose cumulative probability exceeds p (nucleus / top-p sampling) and rescaled by a temperature T. Sampling trades a little adequacy for diversity. The distinction matters twice in this work: (i) for *generating* dialect we want some diversity, and (ii) for *back-translation* we deliberately sample so that synthetic sources are not the artificially regular outputs of beam search (Edunov et al., 2018).

3.5 Back-translation

Back-translation (Sennrich et al., 2016b; Edunov et al., 2018) exploits abundant target-side data when parallel data is scarce. To build augmented pairs for a source→target direction, one runs a *reverse* model (target→source) over real target sentences to produce synthetic sources, then trains on the (synthetic-source, real-target) pairs. Because the target is real, the model never learns to produce machine artefacts; because the source is only an input, its imperfections are tolerable. For our standard→dialect task the "target" is real dialect, so we run the dialect→standard model to hallucinate standard sources—and, following Edunov et al. (2018), we *sample* rather than beam-decode them.

3.6 The one-to-many problem and its evaluation challenge

Normalization is many-to-one: many dialectal spellings collapse to one standard form, so a single reference is (nearly) the only correct answer and reference-based overlap metrics are meaningful. Generation is the inverse, one-to-many: one standard sentence has many equally valid dialectal realizations (different markers, pronouns, and orthographic choices), and the corpus supplies only one. A model that produces a perfectly idiomatic alternative to the reference is penalized by BLEU exactly as a wrong answer would be, and—because dialectal rewrites change only a few tokens—a model that simply *copies* the standard input can score deceptively high on overlap metrics while performing no transfer at all. This is the concrete form of the warning in Edunov et al. (2018) that n-gram overlap correlates poorly with stylistic quality and diversity. It motivates the dedicated metrics of Section 7: measures of whether the *right kind* of change was made, independent of matching one particular reference, plus an explicit copy-rate to expose under-generation.

4 Datasets

ViDia2Std (Features A and C). ViDia2Std (Ta et al., 2026) provides 13,657 dialect/standard sentence pairs mined from public Facebook comments across all 63 Vietnamese provinces and labelled by native annotators with one of three regions (Northern, Central, Southern). We use the corpus in *both* directions: read forward (dialect→standard) it supplies Feature A; read in reverse (standard→dialect) the same pairs, retagged with a region-specific prefix, supply the novel Feature C. The corpus is severely region-imbalanced (Central ≈66%, Southern ≈27%, Northern ≈7%; Figure 1), which directly motivates the rebalancing and back-translation of Section 5.

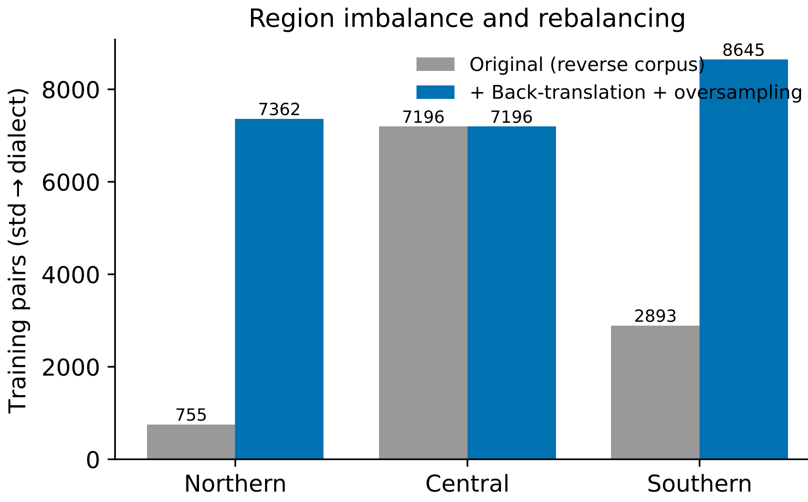


Figure 1: Regional imbalance in ViDia2Std: Central dialect dominates while Northern is severely under-represented, motivating oversampling and targeted back-translation.

ViLexNorm (Feature B). ViLexNorm (Nguyen et al., 2024) provides 10,467 lexical-normalization pairs (teencode/social-media text → standard Vietnamese) with an 8:1:1 train/dev/test split. Including it as a third task exposes the shared encoder to a broad range of non-standard spellings, which we expect to help the dialect tasks that face the same orthographic noise.

Unified format. All datasets are converted to a common text-to-text format: each example has a task condition, a source sentence, and a target sentence. The condition selects one of the five mappings: dialect-to-standard normalization, lexical normalization, or standard-to-dialect generation for one of the three regions. In implementation, this condition is realized as a short prefix attached to the source sentence. Table 1 reports the resulting split sizes, and Figure 2 shows the overall task composition.

Table 1: Per-task example counts in the processed splits.

Task	Train	Dev	Test
Dialect-to-standard normalization	10,844	1,181	1,597
Central standard-to-dialect generation	7,196	767	1,066
Southern standard-to-dialect generation	2,893	319	330
Northern standard-to-dialect generation	755	95	201
Lexical normalization	8,372	1,050	1,045
Total	30,060	3,412	4,239

5 Method

In this section we describe how a single model is trained to perform all five task variants. We cover the shared architecture and task-prefix conditioning (Section 5.1), the construction of the reverse corpus that makes standard-to-dialect generation possible (Section 5.2), the sampling-based back-translation that augments data-starved regions (Section 5.3), region rebalancing (Section 5.4), and the transfer-aware criterion used for model selection (Section 5.5).

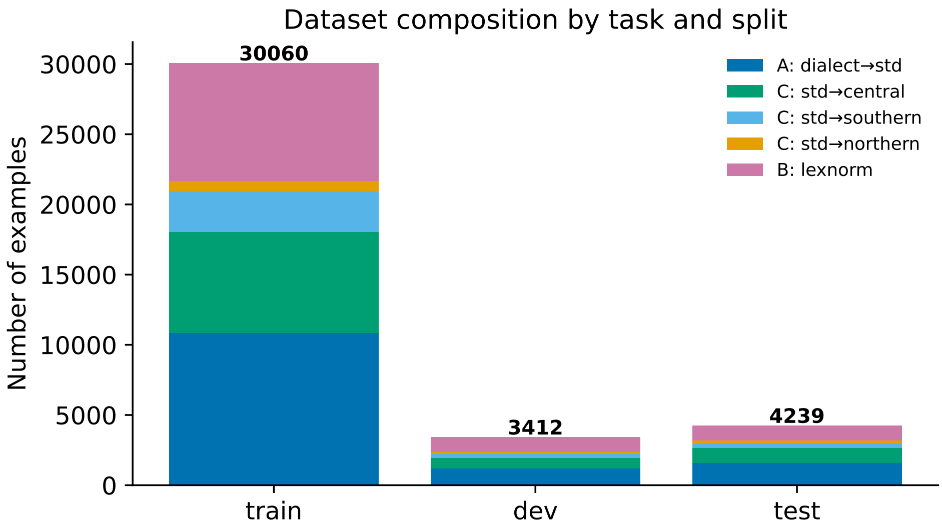


Figure 2: Composition of the unified multi-task training set across the five task prefixes (three regions of Feature C, plus Features A and B).

5.1 Architecture and multi-task task-prefix conditioning

We fine-tune a single BARTpho-syllable model (Tran et al., 2022)—an mBART-style encoder–decoder (Liu et al., 2020) operating on Vietnamese syllable tokens—for all tasks in a unified text-to-text format. Each example is associated with a task condition c, indicating one of five mappings: dialect-to-standard normalization, lexical normalization, or standard-to-dialect generation for the Northern, Central, and Southern regions. The implementation realizes this condition as a short textual prefix prepended to the source sentence, following the text-to-text recipe of Raffel et al. (2020).

For an input sentence $x = (x_1, \ldots, x_m)$, target sentence $y = (y_1, \ldots, y_n)$, and task condition c, the encoder receives the prefixed source

$$\tilde{x} = [p_c;\, x], \tag{3}$$

where p_c is the prefix associated with task c. The decoder then defines the conditional distribution

$$P_\theta(y \mid x, c) = \prod_{t=1}^{n} P_\theta(y_t \mid y_{<t}, \tilde{x})\,. \tag{4}$$

No task-specific parameters, embeddings, or prediction heads are introduced. Thus, the same encoder, decoder, and tokenizer are shared across all tasks, while the prefix provides the only signal that distinguishes the desired mapping.

Given a multi-task training set $\widetilde{\mathcal{D}}$, the model is trained with the teacher-forced negative log-likelihood

$$\mathcal{L}(\theta) = -\frac{1}{|\widetilde{\mathcal{D}}|} \sum_{(x,y,c)\in\widetilde{\mathcal{D}}} \sum_{t=1}^{|y|} \log P_\theta(y_t \mid y_{<t}, \tilde{x})\,. \tag{5}$$

This objective is identical across the three features; only the prefix and the source–target direction change.

5.2 Reverse-corpus construction for Feature C

ViDia2Std is originally distributed as dialect-to-standard parallel data. We write each example as (d_i, s_i, r_i), where d_i is the dialect sentence, s_i is the corresponding standard sentence, and r_i

is the annotated dialect region. The normalization direction uses the pair in its original order:

$$d_i \longrightarrow s_i. \tag{6}$$

To construct Feature C, we reverse the same pair and condition the model on the region label:

$$s_i \longrightarrow d_i \qquad \text{conditioned on } r_i. \tag{7}$$

Thus, the standard sentence becomes the source, the original human-written dialect sentence remains the target, and the region determines which standard-to-dialect prefix is attached to the input. This gives a genuinely parallel training signal for region-conditioned generation without introducing machine-generated dialect targets.

Lexical normalization is added in the same text-to-text format. Its examples map noisy social-media or teencode input to standard Vietnamese:

$$x_j^{\text{noise}} \longrightarrow y_j^{\text{std}}. \tag{8}$$

All three features can therefore be optimized with the same sequence-to-sequence objective in Eq. 5.

5.3 Sampling-based back-translation for data-starved regions

The reversed ViDia2Std corpus inherits the original regional imbalance: Central has many more examples than Southern and especially Northern. To increase the amount of training data for the data-starved regions, we apply back-translation (Sennrich et al., 2016b; Edunov et al., 2018).

For a real dialect sentence d from region r, we first use the dialect-to-standard model to generate a synthetic standard source $\hat{s}$:

$$\hat{s} \sim P_{\theta_A}(s \mid d), \tag{9}$$

where θ_A denotes the normalization model. We then form an additional standard-to-dialect training pair

$$\hat{s} \longrightarrow d \qquad \text{conditioned on } r. \tag{10}$$

The important asymmetry is that only the source side is synthetic; the target side remains real dialectal text. This prevents the generator from learning to imitate machine-produced dialect.

Following Edunov et al. (2018), we sample the synthetic standard source rather than relying on deterministic beam search. Sampling produces more varied sources and reduces the risk of training on overly regular synthetic inputs. A generated pair is retained only if it passes the filtering function

$$Q(\hat{s}, d, r) = Q_{\text{rt}}(\hat{s}, d) \cdot Q_{\text{len}}(\hat{s}, d) \cdot Q_{\text{mark}}(d, r), \tag{11}$$

where Q_{rt} denotes round-trip agreement, Q_{len} enforces a length-ratio constraint, and Q_{mark} checks that the real dialect target contains at least one marker associated with the intended region. The filtered back-translated set for region r is

$$\mathcal{B}_r = \{(\hat{s}, d, r) : Q(\hat{s}, d, r) = 1\}. \tag{12}$$

This augmentation is concentrated on Northern and Southern, where the additional training signal is most useful.

5.4 Region rebalancing

Back-translation is combined with explicit rebalancing of the standard-to-dialect training data. Let $\mathcal{C}_r$ be the original reversed corpus for region r, and let α_r be its oversampling factor. The effective standard-to-dialect training set is

$$\widetilde{\mathcal{C}} = \bigcup_{r\in\{\mathrm{N},\mathrm{C},\mathrm{S}\}} \left(\alpha_r\mathcal{C}_r \cup \mathcal{B}_r\right), \tag{13}$$

where $\alpha_r\mathcal{C}_r$ denotes repeated copies of examples from region r. Regions with fewer original examples receive larger effective weight. The full multi-task training set is then

$$\widetilde{\mathcal{D}} = \mathcal{D}_{\mathrm{norm}} \cup \mathcal{D}_{\mathrm{lex}} \cup \widetilde{\mathcal{C}}, \tag{14}$$

where $\mathcal{D}_{\mathrm{norm}}$ is the dialect-to-standard normalization data and $\mathcal{D}_{\mathrm{lex}}$ is the lexical-normalization data. Oversampling is applied at data-build time rather than through a dynamic sampler, which keeps the effective epoch length explicit and the training mixture reproducible.

5.5 Transfer-aware model selection

Validation BLEU alone is not a reliable selection criterion for standard-to-dialect generation. Since many dialectal rewrites differ from the standard input by only a few tokens, a model that copies the source can obtain high overlap scores while making little or no dialectal transfer. We therefore select checkpoints using a transfer-aware validation score.

Let $S_c(\theta)$ be the validation score for task condition c. For normalization tasks we use the usual overlap-based score, while for standard-to-dialect generation we use Dialectal Edit-Recall:

$$S_c(\theta) = \begin{cases} \mathrm{BLEU}_c(\theta)/100, & \text{if } c \text{ is a normalization task,} \\ \mathrm{EditRecall}_c(\theta), & \text{if } c \text{ is a standard-to-dialect task.} \end{cases} \tag{15}$$

The selected checkpoint is

$$\theta^\star = \arg\max_{\theta\in\mathcal{C}} \frac{1}{|\mathcal{G}|} \sum_{c\in\mathcal{G}} S_c(\theta), \tag{16}$$

where $\mathcal{C}$ is the set of saved checkpoints and $\mathcal{G}$ is the set of validation task conditions. This macro-average gives each task equal influence and prevents the selected checkpoint from being dominated by high-overlap copying behaviour.

6 Baselines

We compare the learned model against four non-neural baselines, all evaluated with the identical metric code of Section 7 so that numbers are directly comparable.

- **Leave-As-Is (LAI / copy).** The output equals the input. This is the crucial floor for one-to-many tasks: because dialectal rewrites touch few tokens, LAI attains high BLEU/chrF despite performing zero transfer (its copy-rate is 1.0 and its Dialectal Edit-Recall is 0 by construction). Any model that does not beat LAI on the *transfer* metrics is not doing the task.
- **Dictionary.** A token-level substitution map learned from the training pairs (most frequent target token for each source token), applied position-wise. It transfers aggressively (copy-rate 0) but ignores context.
- **Rule-based.** Hand-crafted, per-region substitution rules over the curated dialectal markers. This is the strongest non-neural baseline on overlap metrics and a useful sanity check on the marker lists.

Table 2: Baseline results per task on the test split (BLEU/chrF on a 0–100 scale; ROUGE-L, copy-rate, DFR, edit-recall in $[0, 1]$; ERR may be negative). The **transfer** column is WER/CER for Dialect-to-standard, DFR/edit-recall for the std→dialect tasks, and ERR for Lexical normalization. Note that Leave-As-Is (copy) scores high BLEU yet edit-recall 0—the copy trap.

Task	Baseline	BLEU	chrF	ROUGE-L	copy	transfer
Dialect-to-standard	LAI	48.55	63.88	0.770	1.000	WER 0.321 / CER 0.195
	Dictionary	3.65	23.31	0.554	0.000	WER 0.683 / CER 0.532
	Rule-based	55.29	69.03	0.828	0.356	WER 0.277 / CER 0.163
	Retrieval	3.99	14.54	0.310	0.000	WER 1.024 / CER 0.798
Central standard-to-dialect	LAI	44.23	61.90	0.739	1.000	DFR 0.308 / edit 0.000
	Dictionary	4.45	24.42	0.586	0.000	DFR 0.495 / edit 0.432
	Rule-based	50.40	67.74	0.810	0.233	DFR 0.651 / edit 0.291
	Retrieval	4.00	14.63	0.312	0.000	DFR 0.305 / edit 0.200
Southern standard-to-dialect	LAI	47.91	65.90	0.763	1.000	DFR 0.598 / edit 0.000
	Dictionary	5.14	25.03	0.547	0.000	DFR 0.508 / edit 0.435
	Rule-based	50.54	67.38	0.776	0.555	DFR 0.677 / edit 0.104
	Retrieval	5.93	16.15	0.316	0.000	DFR 0.365 / edit 0.192
Northern standard-to-dialect	LAI	74.98	86.43	0.942	1.000	DFR 0.740 / edit 0.000
	Dictionary	23.20	44.25	0.718	0.000	DFR 0.707 / edit 0.468
	Rule-based	76.73	87.26	0.944	0.537	DFR 0.890 / edit 0.154
	Retrieval	2.66	13.26	0.286	0.000	DFR 0.455 / edit 0.148
Lexical normalization	LAI	68.29	78.73	0.847	1.000	ERR 0.000
	Dictionary	6.92	26.48	0.545	0.000	ERR −2.088
	Rule-based	68.29	78.73	0.847	1.000	ERR 0.000
	Retrieval	4.93	14.24	0.290	0.000	ERR −2.781

- **Retrieval.** For a query, return the target of its nearest training *source* under SBERT sentence embeddings. This tests how far pure memorization of the training set can go.

Table 2 reports the four baselines per task. The decisive reading is the *copy trap*: Leave-As-Is posts a high BLEU on the high-overlap dialect tasks (e.g. 74.98 on Northern, where the dialect is orthographically closest to standard), yet its Dialectal Edit-Recall is exactly 0—it performs no transfer at all. The Rule-based system is the strongest non-neural method on overlap but still recalls only a small fraction of the gold substitutions (0.10–0.29 across regions), while the Dictionary transfers aggressively (copy-rate 0) at a catastrophic cost to fluency (BLEU below 6) and the Retrieval baseline is uniformly weak. The improved model (Section 8) beats *every* baseline on Edit-Recall and DFR *while also* clearing the copy BLEU floor. Figure 3 visualizes the comparison.

7 Evaluation Metrics

We use overlap and semantic metrics where they are valid (normalization) and add transfer-specific metrics where they are not (generation). Unless noted, BLEU and chrF are reported on a 0–100 scale and the remaining quantities in $[0, 1]$.

BLEU. Corpus BLEU (Papineni et al., 2002) is the geometric mean of modified n-gram precisions ($n = 1 \ldots 4$) times a brevity penalty BP:

$$\text{BLEU} = BP \cdot \exp\Big(\textstyle\sum_{n=1}^{4} w_n \log p_n \Big), \qquad BP = \min\big(1,\, e^{1-r/h}\big), \tag{17}$$

with p_n the clipped n-gram precision, $w_n = \frac{1}{4}$, h the hypothesis length and r the reference length. We compute it with SacreBLEU (Post, 2018) for reproducibility. It is appropriate for the many-to-one normalization direction but weak for one-to-many generation (Section 3.6).

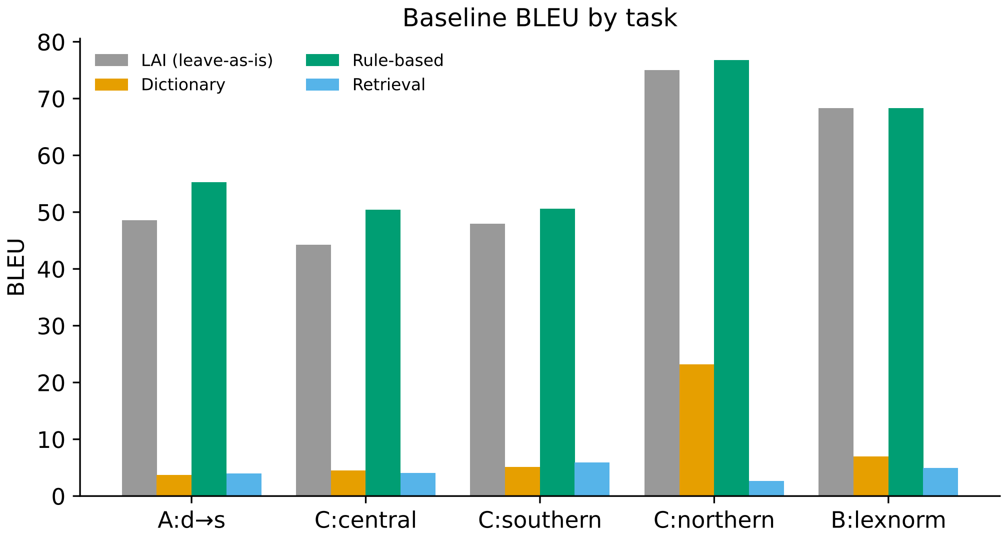


Figure 3: Baseline comparison across tasks. The contrast between high overlap and low transfer for the copy baseline is the key reading.

chrF. chrF (Popović, 2015) is an F-score over character n-grams (we use the chrF++ variant that also includes low-order word n-grams). Operating on characters makes it robust to tokenization and to the rich diacritics of Vietnamese.

ROUGE-L. ROUGE-L (Lin, 2004) is the F-measure of the longest common subsequence between hypothesis and reference, rewarding in-order content overlap without requiring contiguity.

METEOR. METEOR (Banerjee and Lavie, 2005) aligns unigrams (exact and stem matches) and computes a recall-weighted harmonic mean with a fragmentation penalty, correlating better with human judgement than raw precision metrics.

WER / CER. For comparability with ViDia2Std (Ta et al., 2026) on the normalization direction we report Word and Character Error Rate—the Levenshtein edit distance between hypothesis and reference, normalized by reference length (lower is better):

$$\mathrm{WER} = \frac{S + D + I}{N_{\mathrm{ref}}^{\mathrm{words}}}, \qquad \mathrm{CER} = \frac{S_c + D_c + I_c}{N_{\mathrm{ref}}^{\mathrm{chars}}}, \tag{18}$$

with S, D, I the substitution, deletion, and insertion counts.

BERTScore. BERTScore (Zhang et al., 2020) matches hypothesis and reference tokens by contextual-embedding cosine similarity and reports the F_1 of the greedy matching, capturing semantic equivalence that surface metrics miss; we use a multilingual model for Vietnamese.

ERR (Error Reduction Rate) — Feature B. For lexical normalization we report the error reduction over the do-nothing (LAI) baseline. Letting $E(\cdot, \cdot)$ count positionally mismatched tokens against the reference,

$$\mathrm{ERR} \;=\; 1 - \frac{E(\hat{y}, y)}{E(x, y)}, \tag{19}$$

where x is the noisy source, $\hat{y}$ the system output, and y the gold normalization. $\mathrm{ERR} = 1$ means all normalization errors were fixed, $\mathrm{ERR} = 0$ matches LAI, and $\mathrm{ERR} < 0$ means the system introduced more errors than it fixed.

DFR (Dialect Feature Recall) — Feature C. DFR measures whether the output contains the region's characteristic dialectal markers that the reference exhibits. Let M_r be the curated marker set for region r, and for an example let $R = \{m \in M_r : m \in y\}$ be the markers present in the reference. The per-example score is the fraction of those markers reproduced by the system output $\hat{y}$:

$$\text{DFR} = \frac{\big|\{m \in R : m \in \hat{y}\}\big|}{|R|}, \qquad |R| > 0, \tag{20}$$

averaged over examples whose reference contains at least one marker, and reported both per region and overall. DFR directly answers "does it sound like the right region?" but depends on a marker list, so we pair it with a marker-independent measure.

Dialectal Edit-Recall (marker-independent). This metric asks whether the model makes the *same token changes* that the gold reference makes relative to the standard source, without reference to any marker list. Let $T(\cdot)$ be the set of (lowercased) tokens of a sentence. The tokens the reference *introduces* over the source are $A_{\text{gold}} = T(y) \setminus T(x)$, and those the model introduces are $A_{\text{sys}} = T(\hat{y}) \setminus T(x)$. Then

$$\text{Recall} = \frac{|A_{\text{gold}} \cap A_{\text{sys}}|}{|A_{\text{gold}}|}, \qquad\qquad \text{Precision} = \frac{|A_{\text{gold}} \cap A_{\text{sys}}|}{|A_{\text{sys}}|}, \tag{21}$$

with F_1 their harmonic mean, averaged over examples with $|A_{\text{gold}}| > 0$. Recall captures how many of the intended dialectal substitutions were made; precision penalizes spurious changes. Because a copy makes no substitutions, Leave-As-Is has Edit-Recall 0 regardless of its BLEU, which is precisely why this metric, and not BLEU, anchors transfer-aware model selection (Section 5.5).

Region accuracy (marker-based). To check controllability, we classify each output to the region whose markers it most contains and report the fraction matching the *requested* region (over outputs containing at least one marker), together with the fraction of outputs that contain no marker of any region (a no-marker / under-generation rate).

Copy-rate. The fraction of outputs identical (case-insensitively) to the input. High copy-rate exposes under-generation and, read alongside Edit-Recall, separates genuine transfer from copying.

8 Results

We report the final multi-task model on the test split. Table 3 collects the headline numbers; Table 4 isolates the gains of our data and selection strategy over the baseline model; the baselines of Table 2 provide the non-neural reference points.

Normalization (Feature A) positioning. On dialect→standard our syllable-level model reaches 82.10 BLEU, 88.44 chrF, and 0.941 ROUGE-L, with WER 0.1120 and CER 0.0695. For context, the ViDia2Std benchmark (Ta et al., 2026) reports its strongest system, mBART-large-50 (611M parameters), at 0.8166 BLEU / 0.9384 ROUGE-L / 0.1226 WER / 0.0754 CER, and their BARTpho-syllable-base (132M) at 0.7597 BLEU. Our 396M-parameter model thus *matches and slightly edges* the much larger SOTA system—its WER (0.1120) and CER (0.0695) are both below mBART-large-50's—which we present as a strong positioning result with fewer parameters rather than a claim of new state of the art. Figure 4 visualizes the per-task numbers.

Table 3: Main results of the multi-task model per task (test split). BLEU/chrF on a 0–100 scale; ROUGE-L, copy-rate, DFR, edit-recall in $[0, 1]$; WER/CER lower is better. ViDia2Std reference numbers are quoted from Ta et al. (2026) (BLEU/ROUGE-L rescaled to the 0–100 / $[0, 1]$ conventions used here).

Task	BLEU	chrF	ROUGE-L	copy	task-metric
Dialect-to-standard (A)	82.10	88.44	0.941	0.010	WER 0.112 / CER 0.069
Central standard-to-dialect (C)	69.09	80.14	0.906	0.005	DFR 0.901 / edit 0.714
Southern standard-to-dialect (C)	70.36	81.74	0.892	0.018	DFR 0.897 / edit 0.646
Northern standard-to-dialect (C)	87.68	93.47	0.977	0.090	DFR 0.897 / edit 0.635
Lexical normalization (B)	87.45	93.09	0.963	0.039	ERR 0.593
ref: mBART-large-50 (A) (Ta et al., 2026)	81.66	—	0.938	—	WER 0.123 / CER 0.075
ref: BARTpho-syll-base (A) (Ta et al., 2026)	75.97	—	—	—	—

Table 4: Improvement of the final model over the baseline model on the four dialect tasks (test split), *baseline→improved*. For Dialect-to-standard the transfer column is WER/CER (lower is better); for the std→dialect tasks it is DFR / Edit-Recall. The headline is **Northern**, the data-starved region targeted by rebalancing and back-translation: Edit-Recall and DFR rise sharply while copy-rate *falls*. Central and Southern are maintained or slightly improved, and Feature A is preserved.

Task	BLEU	DFR / WER	Edit-Rec / CER	copy-rate
Dialect-to-standard (A)	81.68 → 82.10	0.1157 → 0.1120	0.0713 → 0.0695	0.010 → 0.010
Central standard-to-dialect (C)	69.51 → 69.09	0.896 → 0.901	0.719 → 0.714	0.008 → 0.005
Southern standard-to-dialect (C)	69.06 → 70.36	0.889 → 0.897	0.617 → 0.646	0.024 → 0.018
Northern standard-to-dialect (C)	**80.16 → 87.68**	**0.817 → 0.897**	**0.385 → 0.635**	**0.209 → 0.090**

Generation (Feature C). On standard→dialect the model attains an overall DFR of about 0.90 (per-region: Central 0.901, Southern 0.897, Northern 0.897) and a marker-independent Dialectal Edit-Recall of 0.714 (Central), 0.646 (Southern), and 0.635 (Northern). Crucially the model's copy-rate is far below the Leave-As-Is floor of 1.0—0.005 on Central, 0.018 on Southern, and 0.090 on Northern (the hardest region)—confirming that the gains come from genuine transfer rather than copying. Region accuracy is high on Central (0.917) but lower on Southern (0.424) and Northern (0.209); as Section 9 explains, this marker-based figure is depressed by overlapping marker inventories rather than by a real failure to transfer. Figure 5a contrasts copy-driven BLEU with model BLEU and Figure 5b breaks DFR down by region.

Lexical normalization (Feature B). On ViLexNorm the model reaches an Error Reduction Rate of 0.593 and 87.45 BLEU, exceeding the 0.5774 ERR of the ViLexNorm BARTpho-syllable result.

Improvement over the baseline model. Table 4 contrasts the baseline model (before targeted rebalancing and back-translation) with the improved model. The gains concentrate almost entirely on *Northern*, the data-starved region that rebalancing and back-translation were designed to help: its Edit-Recall rises from 0.385 to 0.635, its copy-rate *falls* from 0.209 to 0.090, its BLEU climbs from 80.2 to 87.7, and its DFR improves from 0.817 to 0.897. Central and Southern are maintained or slightly improved—Southern Edit-Recall rises from 0.617 to 0.646 and DFR from 0.889 to 0.897, while Central moves within noise—and Feature A normalization is preserved (BLEU 81.68 → 82.10, WER 0.1157 → 0.1120). The improvement is exactly where the data intervention was aimed, with no regression elsewhere.

9 Analysis and Error Taxonomy

We organize the analysis around five error modes of the generation direction and the metric that detects each: (i) *under-transfer / copying* (copy-rate, Edit-Recall); (ii) *wrong-region leakage*

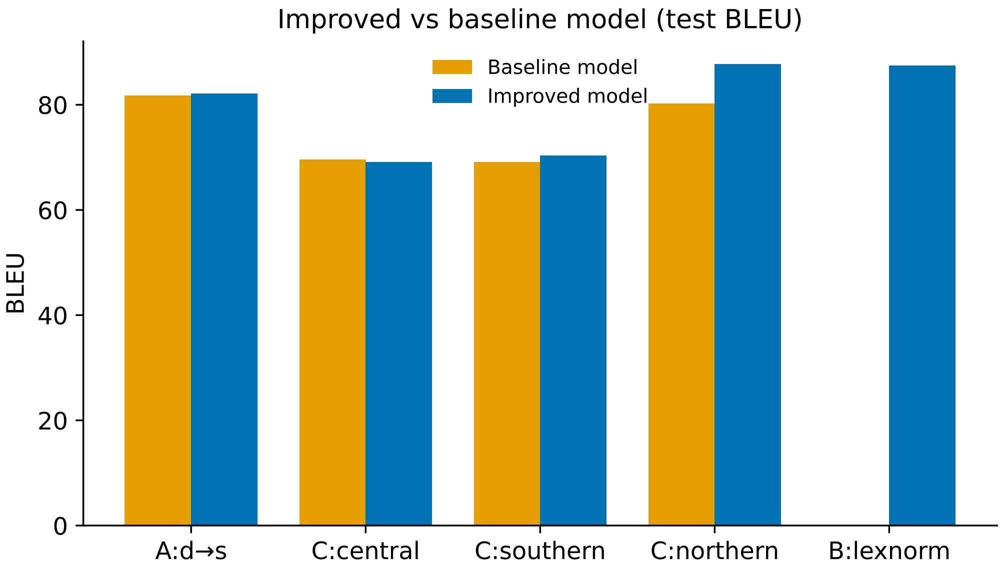
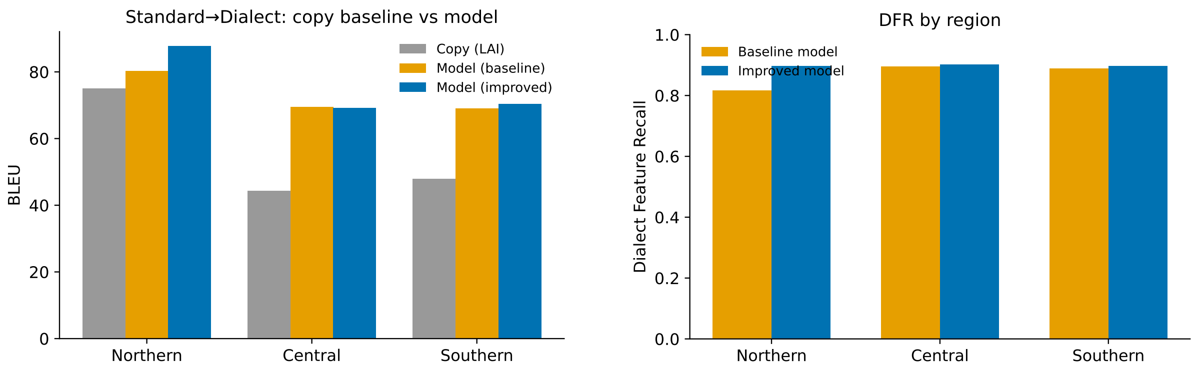


Figure 4: Main multi-task results across the five task variants.



(a) Copy (LAI) BLEU vs. model BLEU.

(b) Dialect Feature Recall by region.

Figure 5: The copy baseline inflates BLEU (left), motivating the transfer metrics (right). *(Figures generated by a separate plotting step.)*

(region accuracy / the confusion matrix); (iii) *over-transfer* (Edit-Precision); (iv) *semantic drift* (BERTScore); and (v) *regional sparsity effects*, concentrated on Northern. The remainder of this section grounds each mode in the model's actual outputs.

9.1 Learned lexical substitutions

Table 5 lists the most frequent standard→dialect token substitutions the model produced on the test set, per region (mined directly from output/source alignment). These are not hand-written rules: the model *induced* them from the parallel data. They are linguistically correct, well-attested regional lexicalisations—Central *này→ni*, *vậy→rứa*, *đâu→mô*, *gì→chi*, *sao→răng*, *tôi→tui*; Northern orthographic/lexical variants *trời→giời*, *mẹ→u/bu*, *nhỉ→nhở*, *chưa→chửa*; and Southern *về→dìa*, *không→hông*, *thật→thiệt*, *xem→coi*, *đắt→mắc*. That the model recovers the canonical function words, pronouns, and kinship terms of each region from data alone is direct evidence that Feature C learns genuine dialect lexicon rather than surface noise.

Table 5: Most frequent learned standard→dialect lexical substitutions per region (token, with output count), mined from the model's test outputs. All are well-attested regional forms.

	Central	Northern	Southern
1	này → ni (15)	nhỉ → nhờ (28)	về → dìa (5)
2	vậy → rứa (10)	nhỉ → nhể (18)	không → hông (5)
3	nào → mô (5)	nhỉ → nhở (12)	thật → thiệt (3)
4	gì → chi (4)	giàu → giầu (11)	xem → coi (3)
5	chứ → chơ (4)	trời → giời (10)	đắt → mắc (3)
6	đâu → mô (4)	chưa → chửa (9)	này → nè (1)
7	sao → răng (3)	mẹ → u (5)	chăn → mền (1)
8	tôi → tui (2)	mẹ → bu (4)	hạt → hột (1)

Table 6: Region-confusion matrix. Rows: requested region; columns: region detected by the heuristic marker classifier in the output ("none" = no marker found). Off-diagonal mass is inflated by overlapping marker inventories and by Northern's orthographic closeness to standard; see text.

Requested ↓ / Detected →	Northern	Central	Southern	none
Northern	91	79	16	15
Central	167	850	39	10
Southern	98	96	126	10

9.2 Region controllability and the confusion matrix

Table 6 gives the region-confusion matrix: rows are the *requested* region, columns are the region whose markers a heuristic detector found in the output (plus a "none" column for outputs with no detected marker). Central is well controlled—850 of 886 marked outputs land on the diagonal. Northern and Southern leak: Northern requests are frequently detected as Central (79) and Southern requests split across Central (96) and Northern (98). This is partly genuine confusion under imbalance, but it is substantially an *artefact of overlapping marker inventories*: several markers (e.g. *mô*, *tui*) belong to more than one region, and Northern dialect is orthographically the closest to standard, so it carries few distinctive markers for the detector to fire on. **For honesty we therefore do not report a raw "wrong-region" error count**—the heuristic marker classifier inflates it through marker overlap—and instead report this confusion matrix together with the marker-independent Dialectal Edit-Recall, which credits the model for making the correct token changes regardless of which region's marker list claims them.

9.3 One-to-many diversity under sampling

To confirm that Feature C produces genuine variation—not a single memorized rewrite—we sample five outputs per standard input and measure Self-BLEU among them (lower means more diverse). Self-BLEU under sampling is 81.6 for Central, 86.7 for Southern, and 93.7 for Northern. Sampling thus yields real variation, most for Central and least for Northern, consistent with Northern being orthographically closest to standard (fewer alternative realizations to sample) and with its data scarcity. For instance, the standard "Tại sao bạn không trả lời tin nhắn của tôi?" yields Central samples ranging over "Mắc mớ gì…", "Răng bạn…", and "Mắc chi…"—distinct, idiomatic realizations that a single-reference BLEU would penalize.

9.4 Tokenization: syllables vs. subwords

The probe in Table 7 makes Section 3.1's robustness argument concrete. Standard syllables tokenize as whole units (`▁Anh`, `▁đi`, `▁đâu`, `▁rồi`), whereas rarer dialectal forms decompose into subwords: *Bả*→`[▁B, ả]`, *rứa*→`[▁r, ứ, a]`, and *Hông*→`[▁H, ông]`. Because every white-space unit remains a valid token sequence, out-of-vocabulary dialect spellings degrade gracefully

Table 7: SentencePiece tokenization of standard vs. dialect surface forms. Standard syllables are whole pieces; rarer dialect forms fragment into subwords (▁ marks a word-initial piece).

Surface form	SentencePiece pieces
"Anh ấy đi đâu rồi?" (std)	`▁Anh ▁ấy ▁đi ▁đâu ▁rồi ?`
"Ảnh đi mô rồi?" (dialect)	`▁Ảnh ▁đi ▁mô ▁rồi ?`
"Bà ấy nói gì vậy?" (std)	`▁Bà ▁ấy ▁nói ▁gì ▁vậy ?`
"Bả nói chi rứa?" (dialect)	`▁B ả ▁nói ▁chi ▁r ứ a ?`
"Không biết" (std)	`▁Không ▁biết`
"Hông biết" (dialect)	`▁H ông ▁biết`

Table 8: Northern std→dialect: baseline (old) vs. improved (new) outputs. The improved model introduces Northern forms (*giả nhời*, *u*) where the baseline copied; the last row shows a still-copied input.

Standard input	Baseline (old)	Improved (new)
Tại sao bạn không trả lời tin nhắn của tôi?	Mắc gì bạn không trả lời tin nhắn của tôi?	Tại sao bạn không *giả nhời* tin nhắn của tôi?
Mẹ tôi nấu ăn rất ngon.	Bu tôi nấu ăn rất ngon.	*u* tôi nấu ăn rất ngon.
Hôm nay trời đẹp quá, chúng ta đi chơi đi.	Hôm nay trời đẹp quá, chúng ta đi chơi đi. *(copied)*	Hôm nay trời đẹp quá, chúng ta đi chơi đi. *(still copied)*

into shorter meaningful pieces rather than triggering a hard segmentation failure—exactly the property that motivates BARTpho-syllable for this task.

9.5 Qualitative examples: Northern before and after

Table 8 shows Northern outputs of the baseline model versus the improved model on the same standard inputs. Where the baseline frequently copied the standard sentence verbatim, the improved model now produces Northern-specific forms: "Tại sao bạn không trả lời…" becomes "…không *giả nhời*…" (the Northern realization of *trả lời*), and "Mẹ tôi…" becomes "*u* tôi…" (the Northern kinship term). We note honestly that Northern remains the hardest region: several inputs (e.g. "Hôm nay trời đẹp quá…", "Anh ấy nói cái gì vậy…") are still copied unchanged, consistent with its 0.090 residual copy-rate and its data scarcity.

A complementary human evaluation of adequacy, fluency, and dialect authenticity is left as a planned human evaluation to validate the automatic transfer metrics.

10 Limitations

Several limitations bound our claims. *Single reference*: ViDia2Std supplies one dialectal realization per sentence, so even our transfer-aware metrics are computed against a single target; the human-eval protocol partly mitigates this but does not eliminate it. *Marker dependence*: DFR and region accuracy rely on a curated marker list and therefore inherit its coverage gaps, which is exactly why we report the marker-independent Edit-Recall alongside them. *Region granularity*: collapsing 63 provinces into three regions hides within-region variation. *Data imbalance*: despite oversampling and targeted back-translation, Northern remains the smallest region and its results are the least reliable. *Synthetic sources*: back-translated standard sources are model output and may carry artefacts, although sampling and keeping real dialect targets limit the damage. *Annotation provenance*: the corpus is drawn from public Facebook comments, with the attendant register, topic, and demographic biases.

11 Conclusion

We presented a single multi-task BARTpho-syllable model that performs dialect→standard normalization, lexical normalization, and—as the novel contribution—region-conditioned standard→dialect generation for the three Vietnamese dialect regions. The work delivers three concrete outcomes. **(A)** On the well-studied normalization direction our 396M-parameter model is competitive with a 611M-parameter SOTA system, reaching 82.10 BLEU at a WER/CER of 0.1120/0.0695—both below ViDia2Std's mBART-large-50—so we match the larger model with roughly a third fewer parameters. **(B)** We built, to our knowledge, the first region-conditioned standard→dialect generator for Vietnamese, reaching an overall DFR of about 0.90 and Edit-Recall of 0.635–0.714; targeted rebalancing and sampling back-translation recover most of the Northern gap (Edit-Recall 0.385 → 0.635, copy-rate 0.209 → 0.090) without regressing the other regions. **(C)** We introduced transfer-aware metrics—Dialect Feature Recall and a marker-independent Dialectal Edit-Recall, paired with copy-rate—that expose the copying behaviour a single-reference BLEU hides and that anchor model selection toward genuine transfer. Together these establish standard→dialect generation as a learnable, measurable task for Vietnamese.

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